Jiarui Li

Phone: 86 - 19801162393 lijiarui@pku.edu.cn
College of Engineering, Peking University, Haidian District, Beijing, China

EDUCATION

Peking University **Beijing, China**
Department of Advanced Manufacturing and Robotics
Bachelor of Engineering in Robotics Engineering ***09/2019 - 07/2024***

- GPA: 3.63/4.0 (86/100)
- Relevant Courses:
  - Practice of Programming in C&C++ (93)
  - Introduction in Computation (A) (91)
  - Set Theory and Graph Theory (91)
  - Electromagnetism (96)
  - Theoretical Mechanics (90)
  - Data Structure and Algorithm (B) (89)
  - An Introduction to Robotics (87.5)
  - Mathematical Analysis (II) (87.5)

RESEARCH INTERESTS

- Robotics
- Reinforcement Learning

RESEARCH EXPERIENCES

The Key Laboratory of Machine Perception, the Speech and Hearing Research Center, the Department of Machine Intelligence (PKU) **Beijing, China**

The research group of Dr. Dingsheng Luo from PKU, working on Artificial Intelligence, Embodied Intelligence, Machine Learning, Speech and Language Processing, Robotics, Humanoid Robot, etc..

I was funded by the President's Fund for Undergraduate Scientific Research to do scientific research projects in the Group.

Undergraduate Research Assistant, Advisor: Prof. Dingsheng Luo ***03/2021- till now***

- **Project: Reinforcement learning based Robot Dual-arm Collaborative Manipulation under Constrained Task Scenarios**
  - Analyze the importance of the problem and the current planning scheme for robot arm operation in constrained task scenarios.
  - Use the "soft constraint" method of reinforcement learning to complete the path planning of the pose constraint under the environment constraint, and make a specific analysis and comparison of the constraint methods, and give a set of solutions.

PROJECTS

PKU **Beijing, China**
Developer ***11/2021- 01/2022***

- **Project: Mechanical Reproduction and Modeling of the "Zhuge Crossbow" of the Three Kingdoms Era in Ancient China Using Solidworks**
  - Collected and analyzed the principle of the implementation of the line generation continuous firing mechanism, and used solidworks to reproduce and improve the "Zhuge Continuous Crossbow" designed by Liang Zhuge in the context of the Three Kingdoms period (220-280).

- Completed the modeling of the Zhuge crossbow using clockwork and modern gear mechanisms as the core of the continuous fire function implementation, designed and determined the specific dimensions of each part, and tested the feasibility of the design on the virtual platform as well as in the real world using 3D printing technology.
- Worked based on Solidworks.

Developer ***10/2021- 12/2021***

- **Project: Modeling of a Wire-Driven Flexible Robotic Arm**
  - Modeled and analyzed the design of a wire-driven snake-like robotic arm, which was composed of multiple nodes. Instead of using motors at each joint, two sets of cables were used as muscles to drive the robot arm.
  - Performed a complete quantitative analysis of the kinematic as well as the static characteristics of this robotic arm using Matlab, and performed specific calculations using forward and inverse dynamics, and also performed some analysis and calculations of the dynamical characteristics using Matlab, and presented the results in graphs in the report.
  - Worked based on Matlab.

Developer ***10/2021- 12/2021***

- **Project: Analyze the Original System and the Closed-loop Feedback System Based on a Second-order Inverted Pendulum Dynamic System**
  - Used Laplace transformations, system functions, and the state space method, and analyzed the original system of the second-order inverted pendulum dynamics as well as the closed-loop feedback system and the situation with different parameters.
  - Used Matlab to represent key elements such as the impulse response, and used these graphs to specify the results, culminating in a full experimental report.
  - Worked based on Matlab.

Developer ***09/2021- 10/2021***

- **Project: Analysis and Synthesis of Electronic Music**
  - Analyzed the composition of electronic music using Fourier series and Fourier transforms, and reproduced a fragment of Beethoven's Fifth Symphony using the results of previous researchers on piano timbres.
  - Analyzed and extracted a guitar performance clip and reconstructed the key elements of the guitar, such as tone and timbre, and used the results to replay a Chinese piece with good results.
  - Worked based on Matlab.

Developer ***11/2020- 01/2021***

- **Project: Modeling and Analysis of Theoretical Problems of Transmission Line Ice Cover Using Matlab Combined with Theoretical Mechanics**
  - Used Matlab to model and analyze the phenomenon of over-icing of transmission lines and its consequences, and used the Lagrangian method to simulate the motion model of transmission lines after breaking.
  - Completed an analysis of de-icing methods represented by de-icing robots, and specifically analyzed the impact of de-icing robots walking on the line on transmission line morphology
  - Worked based on Matlab.

Developer ***04/2020- 06/2020***

- **Project: Game Intelligence Algorithm Design and Competition of Chinese National Standard Mahjong based on C++ Language**
  - Developed an intelligent game algorithm of Chinese national standard mahjong based on C++ language, and continuously improved the performance after four rounds of version iterations, and won a relatively good result in the algorithm competition.
  - Worked based on C++ language.

SKILLS

- **Programming Languages:** Proficient in C Language, C++, Python, Matlab.
- **Software Tools:** Proficient in Solidworks, ROS techniques, gazebo, Matlab, etc.